High-Dimensional Linear Regression: Ridge and Lasso

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1 High-Dimensional Linear Regression

In many modern applications, the number of available predictors can be very large. Examples include genomics, text analysis, recommender systems, finance, and many machine learning problems.

Despite their simplicity, linear models often remain a first method of choice. They are computationally efficient, easy to interpret, and frequently provide surprisingly good predictive performance.

When the number of predictors is large, however, ordinary least squares can perform poorly. The fitted model may have high variance, leading to unstable predictions. Moreover, it is often unreasonable to believe that every predictor is equally important.

A common assumption is that only a relatively small number of predictors have a substantial effect on the response, while many others have little or no influence. Even when this assumption is not exactly true, it is often the case that a few predictors are much more important than the rest.

This suggests that we should fit a linear model while simultaneously discouraging the coefficients of unimportant variables from becoming large. In other words, we would like to *shrink* the regression coefficients of unimportant variables toward zero.

Shrinkage serves two purposes. First, it reduces the variance of the fitted model, often improving predictive accuracy. Second, it helps identify the predictors that are most relevant to the response.

Two of the most important shrinkage methods are:

- **Ridge regression**, which shrinks all coefficients continuously toward zero;
- **The lasso**, which can shrink some coefficients all the way to zero and therefore performs automatic variable selection.

These methods form the foundation of modern high-dimensional linear regression and illustrate the broader principle that introducing a small amount of bias can dramatically reduce variance and improve prediction.

1.1 Ridge Regression

The following slides from Hastie's lecture notes provide an excellent introduction to ridge regression and shrinkage.

Shrinkage Methods

Ridge regression and *Lasso*

- The subset selection methods use least squares to fit a linear model that contains a subset of the predictors.
- As an alternative, we can fit a model containing all p predictors using a technique that *constrains* or *regularizes* the coefficient estimates, or equivalently, that *shrinks* the coefficient estimates towards zero.
- It may not be immediately obvious why such a constraint should improve the fit, but it turns out that shrinking the coefficient estimates can significantly reduce their variance.

Ridge regression

- Recall that the least squares fitting procedure estimates $\beta_0, \beta_1, \ldots, \beta_p$ using the values that minimize

$$\mathrm{RSS} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 .$$

- In contrast, the ridge regression coefficient estimates $\hat{\beta}^R$ are the values that minimize

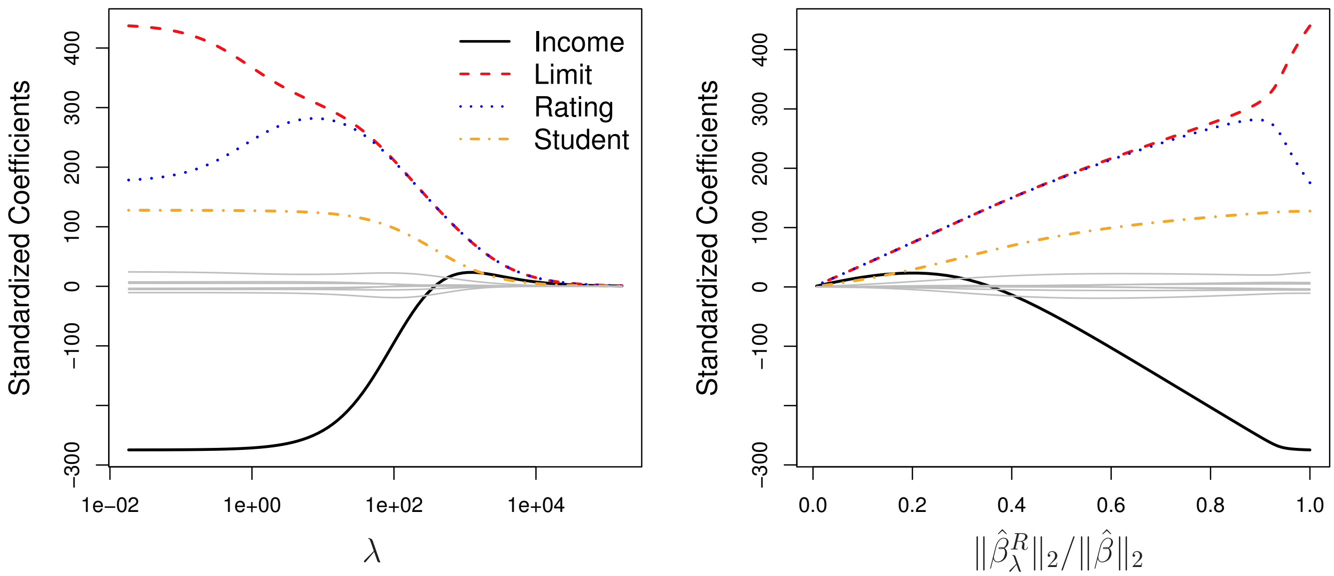
$$\sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 = \mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

where $\lambda \geq 0$ is a *tuning parameter*, to be determined separately.

Ridge regression: continued

- As with least squares, ridge regression seeks coefficient estimates that fit the data well, by making the RSS small.
- However, the second term, $\lambda \sum_j \beta_j^2$, called a *shrinkage penalty*, is small when $\beta_1, \ldots, \beta_p$ are close to zero, and so it has the effect of *shrinking* the estimates of β_j towards zero.
- The tuning parameter λ serves to control the relative impact of these two terms on the regression coefficient estimates.
- Selecting a good value for λ is critical; cross-validation is used for this.

Credit data example



Details of Previous Figure

- In the left-hand panel, each curve corresponds to the ridge regression coefficient estimate for one of the ten variables, plotted as a function of λ.
- The right-hand panel displays the same ridge coefficient estimates as the left-hand panel, but instead of displaying λ on the x-axis, we now display $\|\hat{\beta}_\lambda^R\|_2/\|\hat{\beta}\|_2$, where $\hat{\beta}$ denotes the vector of least squares coefficient estimates.
- The notation $\|\beta\|_2$ denotes the ℓ_2 norm (pronounced "ell 2") of a vector, and is defined as $\|\beta\|_2 = \sqrt{\sum_{j=1}^p {\beta_j}^2}$.

Ridge regression: scaling of predictors

- The standard least squares coefficient estimates are *scale equivariant*: multiplying X_j by a constant c simply leads to a scaling of the least squares coefficient estimates by a factor of $1/c$. In other words, regardless of how the jth predictor is scaled, $X_j\hat{\beta}_j$ will remain the same.
- In contrast, the ridge regression coefficient estimates can change *substantially* when multiplying a given predictor by a constant, due to the sum of squared coefficients term in the penalty part of the ridge regression objective function.
- Therefore, it is best to apply ridge regression after *standardizing the predictors*, using the formula

$$\tilde{x}_{ij} = \frac{x_{ij}}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \overline{x}_j)^2}}$$

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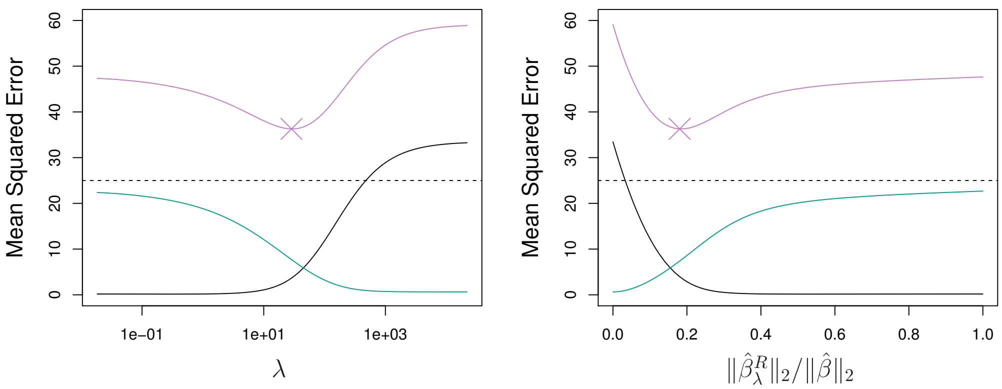
Why Does Ridge Regression Improve Over Least Squares?

The Bias-Variance tradeoff



Simulated data with $n = 50$ *observations,* $p = 45$ *predictors, all having nonzero coefficients. Squared bias (black), variance (green), and test mean squared error (purple) for the ridge regression predictions on a simulated data set, as a function of* λ *and* $\|\hat{\beta}^R_\lambda\|_2/\|\hat{\beta}\|_2$*. The horizontal dashed lines indicate the minimum possible MSE. The purple crosses indicate the ridge regression models for which the MSE is smallest.*

1.2 Choosing the Tuning Parameter

Ridge regression introduces a tuning parameter λ that controls the amount of shrinkage.

When $\lambda = 0$, ridge regression reduces to ordinary least squares. As λ increases, the coefficients are shrunk more aggressively toward zero. A very small value of λ may lead to overfitting, while a very large value may lead to underfitting.

The key practical question is therefore: *how should we choose* λ*?*

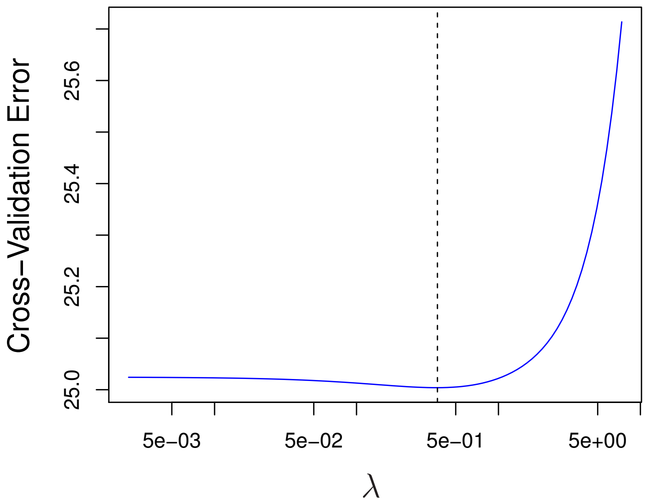
The most common approach is cross-validation. We select a grid of candidate values of λ, fit the ridge regression model for each value, and estimate the corresponding test error using K-fold cross-validation. The value of λ with the smallest estimated cross-validation error is then selected. Finally, ridge regression is re-fit to the entire data set using this chosen value of λ.

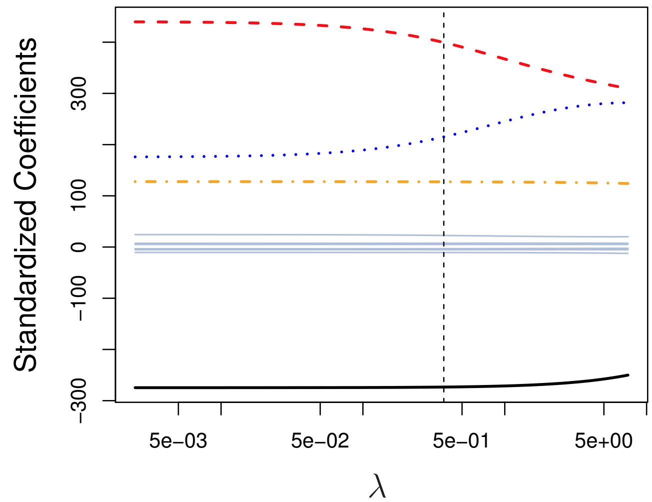
The following slides from Hastie's notes illustrate this procedure on the Credit data set.

Selecting the Tuning Parameter for Ridge Regression and Lasso

- As for subset selection, for ridge regression and lasso we require a method to determine which of the models under consideration is best.
- That is, we require a method selecting a value for the tuning parameter λ or equivalently, the value of the constraint s.
- *Cross-validation* provides a simple way to tackle this problem. We choose a grid of λ values, and compute the cross-validation error rate for each value of λ.
- We then select the tuning parameter value for which the cross-validation error is smallest.
- Finally, the model is re-fit using all of the available observations and the selected value of the tuning parameter.

Credit data example





Left: *Cross-validation errors that result from applying ridge regression to the* `Credit` *data set with various values of* λ*.*
Right: *The coefficient estimates as a function of* λ*. The vertical dashed lines indicates the value of* λ *selected by cross-validation.*

2 Ridge Regression from a Bayesian Viewpoint

We have a response variable y and a single covariate x. In our example, y denotes weekly earnings and x denotes years of experience. The covariate x takes the values $0, 1, \ldots, m$ for some fixed integer m.

Our data is $(x_i, y_i), i = 1, \ldots, n$. We consider the model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \operatorname{ReLU}(x_i - 1) + \cdots + \beta_m \operatorname{ReLU}(x_i - (m-1)) + \epsilon_i \tag{1}$$

where $\operatorname{ReLU}(u) := \max(u, 0)$, and $\epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$. Note we did not include $\operatorname{ReLU}(x_i - m)$ because it always equals 0.

This linear regression equation can be written in the usual notation as:

$$y = X\beta + \epsilon \tag{2}$$

where $y = (y_1, \ldots, y_n)^T$ and X is the $n \times (m+1)$ matrix with columns 1, x, $\operatorname{ReLU}(x - j)$ for $j = 1, \ldots, m-1$.

2.1 Recap: Linear Regression with default priors

As we saw before, the default prior on $\beta_0, \ldots, \beta_m$ in the linear regression model (2) is:

$$\beta_0, \ldots, \beta_m \overset{\text{i.i.d}}{\sim} \text{uniform}(-C, C) \tag{3}$$

with $C \to \infty$. With this prior, the posterior of β given σ becomes

$$\beta \mid \text{data}, \sigma \sim N\left((X^TX)^{-1}X^Ty, \sigma^2(X^TX)^{-1}\right) \tag{4}$$

which means that the posterior mean of β is just the least squares estimator $(X^TX)^{-1}X^Ty$ (note that this posterior mean of β does not depend on σ, so is both the conditional posterior mean given σ, as well as the unconditional posterior mean).

The fact that the posterior mean of β coincides with the least squares estimator is shared by some priors other than (3) as well. For example, this is also true for the following prior:

$$\beta_0, \ldots, \beta_m \overset{\text{i.i.d}}{\sim} N(0, C) \tag{5}$$

as $C \to \infty$. This can be seen by the following general result :

$$\beta \mid \sigma \sim N(0, Q) \implies \beta \mid \text{data}, \sigma \sim N\left(\left(\frac{X^TX}{\sigma^2} + Q^{-1}\right)^{-1} \frac{X^Ty}{\sigma^2}, \left(\frac{X^TX}{\sigma^2} + Q^{-1}\right)^{-1}\right). \tag{6}$$

Exercise: Show this!

From here, it is clear that if Q is chosen so that $Q^{-1} \approx 0$, then we get the same posterior as (4). The prior (5) corresponds to $Q = CI$ so that $Q^{-1} = (1/C)I$ which goes to zero when $C \to \infty$. So the prior (5) leads to the same posterior (4).

2.2 Different Prior for Relu Coefficients

In the earnings-experience regression example, we have seen in class that the least squares estimator for (2) is not interpretable. We want the fitted curve

$$x \mapsto \hat{\beta}_0 + \hat{\beta}_1 x + \sum_{j=2}^{m} \hat{\beta}_j \operatorname{ReLU}(x - (j-1))$$

to be smooth for it to be interpretable. It is reasonable to think that most of the beta's when $j = 2, \dots, m$ should be small. This can be achieved in the Bayesian setting if we use the prior:

$$\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C) \quad \text{and} \quad \beta_2, \dots, \beta_m \overset{\text{i.i.d}}{\sim} N(0, \tau^2). \tag{7}$$

for some small τ. The above prior is equivalent to $\beta \sim N(0, Q)$ where Q is the $(m+1) \times (m+1)$ diagonal matrix with diagonal entries $C, C, \tau^2, \dots, \tau^2$. So using the formula (6), the posterior mean corresponding to this prior is:

$$(\beta \mid \text{data}, \sigma, \tau) = \left(\frac{X^T X}{\sigma^2} + Q^{-1}\right)^{-1} \frac{X^T y}{\sigma^2}.$$

Because Q is diagonal with diagonal entries $C, C, \tau^2, \dots, \tau^2$, it is easy to see that Q^{-1} is also diagonal with diagonal entries $1/C, 1/C, \tau^{-2}, \dots, \tau^{-2}$. Because C is large, we can approximate Q^{-1} by the matrix with diagonal entries $0, 0, \tau^{-2}, \dots, \tau^{-2}$. In other words:

$$Q^{-1} \approx \frac{1}{\tau^2} J \quad \text{where } J = \begin{pmatrix} 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 1 & 0 & \cdot & \cdot & \cdot & 0 \\ 0 & 0 & 0 & 1 & \cdot & \cdot & \cdot & 0 \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & 1 \end{pmatrix}$$

Note that J is an $(m+1) \times (m+1)$ diagonal matrix. Thus the posterior mean becomes:

$$(\beta \mid \text{data}, \sigma, \tau) = \left(\frac{X^T X}{\sigma^2} + \frac{J}{\tau^2}\right)^{-1} \frac{X^T y}{\sigma^2} = \left(X^T X + \frac{J}{\tau^2/\sigma^2}\right)^{-1} X^T y.$$

We first note that the posterior mean now depends on σ and τ only through the ratio $\gamma^2 = \tau^2/\sigma^2$. This gives

$$(\beta \mid \text{data}, \sigma, \tau) = \left(X^T X + \frac{J}{\gamma^2}\right)^{-1} X^T y. \tag{8}$$

It can be verified that this quantity $(X^T X + J/\gamma^2)^{-1} X^T y$ coincides with a ridge regularized least squares estimator. This is shown in the next section.

2.3 Closed Form Solution of Ridge Regression

For the model (1), the ridge regression estimator Ridge for β is given by the minimizer of:

$$\begin{aligned}&\sum_{i=1}^{n}(y_i-\beta_0-\beta_1 x_i-\beta_2\,\mathrm{ReLU}(x_i-1)-\cdots-\beta_m\,\mathrm{ReLU}(x_i-(m-1)))^2\\&+\lambda\left(\beta_2^2+\beta_3^2+\cdots+\beta_m^2\right).\end{aligned} \tag{9}$$

Remark 2.1. *There is no sense in penalizing the linear function part in this case. For general predictors, we can penalize all β's.*

The objective function above can also be written as

$$\|y-X\beta\|^2+\lambda\sum_{j=2}^{m}\beta_j^2.$$

It turns out that Ridge can be written in closed form using matrix notation. To see this, note first that the gradient of the above objective function with respect to β is given by

$$\nabla\left(\|y-X\beta\|^2+\lambda\sum_{j=2}^{m}\beta_j^2\right)=-2X^Ty+2X^TX\beta+2\lambda\begin{pmatrix}0\\0\\\beta_2\\\cdot\\\cdot\\\cdot\\\beta_m\end{pmatrix}$$

Recalling that J is the $(m+1)\times(m+1)$ diagonal matrix with diagonal entries $0,0,1,\dots,1$, we can write

$$\nabla\left(\|y-X\beta\|^2+\lambda\sum_{t=2}^{n-1}\beta_j^2\right)=-2X^Ty+2X^TX\beta+2\lambda\begin{pmatrix}0\\0\\\beta_2\\\cdot\\\cdot\\\cdot\\\beta_{n-1}\end{pmatrix}=-2X^Ty+2X^TX\beta+2\lambda J\beta.$$

Setting this gradient equal to zero, we get

$$-2X^Ty+2X^TX\beta+2\lambda J\beta=0\implies\left(X^TX+\lambda J\right)\beta=X^Ty.$$

which gives

$$\mathrm{Ridge}=(X^TX+\lambda J)^{-1}X^Ty. \tag{10}$$

Note that $\lambda=0$ corresponds to least squares, and $\lambda\to\infty$ leads to linear regression. When λ is set to be neither very close to 0 nor very large, we should get a smooth estimate that still gives a good fit to the data.

Remark 2.2. *Everything we say here in this section can be carried out for general predictors. We can penalize all coefficients in the general case, then J would be replaced by the identity matrix.*

Comparing (10) to (8), it is clear that they are identical when $\lambda=1/\gamma^2$. In other words, when we use the prior (7), then the posterior mean (given τ and σ) coincides with ridge regularized least squares with regularization parameter $\lambda=1/\gamma^2=\sigma^2/\tau^2$.

2.4 Bayesian inference for γ and σ

The big advantage of Bayesian inference is that it also allows inference on γ and σ. This is in contrast to frequentist inference where we have to use cross validation to set λ. Also it is not clear how to estimate σ well in high dimensional settings.

We shall use the prior:

$$\log \gamma, \log \sigma \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty).$$

Recall again that $\tau = \gamma\sigma$. The joint prior on all the parameters (β, γ, σ) is:

$$f_{\beta,\gamma,\sigma}(\beta, \gamma, \sigma) \propto \frac{1}{\gamma\sigma} \frac{1}{\sqrt{\det Q}} \exp\left(-\frac{1}{2}\beta^T Q^{-1} \beta\right).$$

Here Q is the $(m+1) \times (m+1)$ diagonal matrix with diagonal entries $C, C, \gamma^2\sigma^2, \ldots, \gamma^2\sigma^2$.

With this prior, we shall argue in the next section that the posterior is given by:

$$\beta \mid \text{data}, \sigma, \gamma \sim N\left(\left(X^T X + \gamma^{-2} J\right)^{-1} X^T y, \sigma^2 \left(X^T X + \gamma^{-2} J\right)^{-1}\right) \tag{11}$$

and

$$\frac{1}{\sigma^2} \mid \text{data}, \gamma \sim \text{Gamma}\left(\frac{n}{2} - 1, \frac{y^T y - y^T X \left(X^T X + \gamma^{-2} J\right)^{-1} X^T y}{2}.\right) \tag{12}$$

and

$$f_{\gamma|\text{data}}(\gamma) \propto \frac{\gamma^{-m} \left|X^T X + \gamma^{-2} J\right|^{-1/2}}{\left(y^T y - y^T X \left(X^T X + \gamma^{-2} J\right)^{-1} X^T y\right)^{(n/2)-1}}.$$

With this, inference can be carried out by first taking a grid of γ values and computing the above posterior (on the logarithmic scale) at the grid points. This posterior can be used to obtain posterior samples of γ. For each sample of γ, we can then sample σ using the distribution (12). Given samples from both γ and σ, we can then sample β using (11).

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2.5 Full Posterior Calculation under the Reparametrization $\tau = \sigma\gamma$

We start from the regression model

$$y = X\beta + \epsilon, \qquad \epsilon \sim N_n(0, \sigma^2 I_n),$$

where X is an $n \times (m+1)$ design matrix and $\beta = (\beta_0, \beta_1, \ldots, \beta_m)^T$.

The likelihood is

$$f(y \mid \beta, \sigma) \propto \sigma^{-n} \exp\left\{-\frac{1}{2\sigma^2} \|y - X\beta\|_2^2\right\}. \tag{13}$$

We want a prior which leaves the intercept and linear part essentially unregularized, while shrinking the nonlinear spline coefficients $\beta_2, \ldots, \beta_m$. Let

$$J = \text{diag}(0, 0, 1, \ldots, 1),$$

so that

$$\beta^T J\beta = \sum_{j=2}^{m} \beta_j^2.$$

Formally, we take

$$\beta_0, \beta_1$$

to have very diffuse Gaussian priors, and

$$\beta_2, \ldots, \beta_m \mid \tau \overset{\text{i.i.d.}}{\sim} N(0, \tau^2).$$

Ignoring constants that do not depend on the parameters, the prior density of β given τ is therefore

$$f(\beta \mid \tau) \propto \tau^{-(m-1)} \exp\left\{-\frac{1}{2\tau^2}\beta^T J\beta\right\}. \tag{14}$$

We now introduce the reparametrization

$$\tau = \sigma\gamma, \qquad \gamma > 0.$$

Equivalently,

$$\gamma = \frac{\tau}{\sigma}.$$

Rather than placing a prior directly on τ, it is more convenient to place a prior $p(\gamma)$ on the ratio $\gamma = \tau/\sigma$. We also use the usual scale-invariant prior

$$p(\sigma) \propto \frac{1}{\sigma}, \qquad \sigma > 0.$$

Thus, in the (σ, γ) parametrization, the joint prior is

$$f(\beta, \sigma, \gamma) \propto p(\gamma)\, \frac{1}{\sigma}\, (\sigma\gamma)^{-(m-1)} \exp\left\{-\frac{1}{2\sigma^2\gamma^2}\beta^T J\beta\right\}. \tag{15}$$

Equivalently,

$$f(\beta, \sigma, \gamma) \propto p(\gamma)\, \gamma^{-(m-1)}\sigma^{-m} \exp\left\{-\frac{1}{2\sigma^2}\gamma^{-2}\beta^T J\beta\right\}. \tag{16}$$

Combining the likelihood (13) with the prior (16), we obtain the joint posterior

$$f(\beta, \sigma, \gamma \mid y) \propto p(\gamma)\, \gamma^{-(m-1)}\sigma^{-n-m} \exp\left\{-\frac{1}{2\sigma^2}\left[\|y - X\beta\|_2^2 + \gamma^{-2}\beta^T J\beta\right]\right\}. \tag{17}$$

Now define

$$K_\gamma = X^T X + \gamma^{-2} J. \tag{18}$$

Then

$$\begin{aligned}\|y - X\beta\|_2^2 + \gamma^{-2}\beta^T J\beta &= y^T y - 2y^T X\beta + \beta^T X^T X\beta + \gamma^{-2}\beta^T J\beta \\ &= y^T y - 2y^T X\beta + \beta^T K_\gamma\beta.\end{aligned}$$

Completing the square gives

$$y^T y - 2y^T X\beta + \beta^T K_\gamma\beta = (\beta - \mu_\gamma)^T K_\gamma(\beta - \mu_\gamma) + A_\gamma, \tag{19}$$

where

$$\mu_\gamma = K_\gamma^{-1} X^T y \tag{20}$$

and

$$A_\gamma = y^T y - y^T X K_\gamma^{-1} X^T y. \tag{21}$$

Therefore, conditional on σ and γ, the posterior density of β is Gaussian:

$$\beta \mid y, \sigma, \gamma \sim N_{m+1}\left(K_\gamma^{-1} X^T y, \sigma^2 K_\gamma^{-1}\right). \tag{22}$$

Next, we integrate out β from (17). Using (19), we have

$$\begin{aligned} f(\sigma, \gamma \mid y) \propto\ & p(\gamma)\, \gamma^{-(m-1)} \sigma^{-n-m} \exp\left\{-\frac{A_\gamma}{2\sigma^2}\right\} \\ & \times \int_{\mathbb{R}^{m+1}} \exp\left\{-\frac{1}{2\sigma^2}(\beta - \mu_\gamma)^T K_\gamma (\beta - \mu_\gamma)\right\} d\beta. \end{aligned}$$

The Gaussian integral is

$$\int_{\mathbb{R}^{m+1}} \exp\left\{-\frac{1}{2\sigma^2}(\beta - \mu_\gamma)^T K_\gamma (\beta - \mu_\gamma)\right\} d\beta = (2\pi\sigma^2)^{(m+1)/2} |K_\gamma|^{-1/2}.$$

Ignoring constants independent of σ and γ, this gives

$$f(\sigma, \gamma \mid y) \propto p(\gamma)\, \gamma^{-(m-1)} |K_\gamma|^{-1/2} \sigma^{-n+1} \exp\left\{-\frac{A_\gamma}{2\sigma^2}\right\}. \tag{23}$$

Thus, conditional on γ,

$$f(\sigma \mid y, \gamma) \propto \sigma^{-n+1} \exp\left\{-\frac{A_\gamma}{2\sigma^2}\right\}, \qquad \sigma > 0. \tag{24}$$

Equivalently, if

$$u = \frac{1}{\sigma^2},$$

then

$$\sigma = u^{-1/2}, \qquad \left|\frac{d\sigma}{du}\right| = \frac{1}{2} u^{-3/2}.$$

Therefore,

$$\begin{aligned} f(u \mid y, \gamma) &\propto f(\sigma = u^{-1/2} \mid y, \gamma) u^{-3/2} \\ &\propto (u^{-1/2})^{-n+1} \exp\left\{-\frac{A_\gamma u}{2}\right\} u^{-3/2} \\ &= u^{(n-1)/2} u^{-3/2} \exp\left\{-\frac{A_\gamma u}{2}\right\} \\ &= u^{(n-4)/2} \exp\left\{-\frac{A_\gamma u}{2}\right\}. \end{aligned}$$

This is a Gamma density with shape

$$\frac{n-2}{2} = \frac{n}{2} - 1$$

and rate $A_\gamma/2$. Hence

$$\frac{1}{\sigma^2} \mid y, \gamma \sim \text{Gamma}\left(\frac{n}{2} - 1, \frac{A_\gamma}{2}\right). \tag{25}$$

Finally, we integrate out σ from (23) to obtain the marginal posterior of γ. From (23),

$$f(\gamma \mid y) \propto p(\gamma)\,\gamma^{-(m-1)}|K_\gamma|^{-1/2} \int_0^\infty \sigma^{-n+1} \exp\left\{-\frac{A_\gamma}{2\sigma^2}\right\} d\sigma.$$

The integral is a constant multiple of

$$A_\gamma^{-(n/2-1)}.$$

To see this explicitly, set

$$\sigma = s\sqrt{A_\gamma}.$$

Then

$$d\sigma = \sqrt{A_\gamma}\, ds$$

and

$$\sigma^{-n+1} = (s\sqrt{A_\gamma})^{-n+1} = s^{-n+1} A_\gamma^{-(n-1)/2}.$$

Therefore

$$\begin{aligned}
\int_0^\infty \sigma^{-n+1} \exp\left\{-\frac{A_\gamma}{2\sigma^2}\right\} d\sigma &= A_\gamma^{-(n-1)/2} A_\gamma^{1/2} \int_0^\infty s^{-n+1} \exp\left\{-\frac{1}{2s^2}\right\} ds \\
&= A_\gamma^{-(n-2)/2} \int_0^\infty s^{-n+1} \exp\left\{-\frac{1}{2s^2}\right\} ds \\
&\propto A_\gamma^{-(n/2-1)}.
\end{aligned}$$

The remaining integral does not depend on γ, so it is absorbed into the normalizing constant. Hence

$$f(\gamma \mid y) \propto p(\gamma)\,\gamma^{-(m-1)}|K_\gamma|^{-1/2} A_\gamma^{-(n/2-1)}. \tag{26}$$

Substituting back the definitions of K_γ and A_γ, we get

$$f(\gamma \mid y) \propto \frac{p(\gamma)\,\gamma^{-(m-1)}\left|X^T X + \gamma^{-2} J\right|^{-1/2}}{\left(y^T y - y^T X \left(X^T X + \gamma^{-2} J\right)^{-1} X^T y\right)^{n/2-1}}. \tag{27}$$

A common default choice is the log-uniform prior

$$p(\gamma) \propto \frac{1}{\gamma} I\{\gamma_{\text{low}} < \gamma < \gamma_{\text{high}}\}. \tag{28}$$

The full posterior sampling scheme is then:

1. Compute the marginal posterior density $f(\gamma \mid y)$ on a grid using (27).
2. Sample γ from this one-dimensional posterior.
3. Given γ, sample

$$\frac{1}{\sigma^2} \mid y, \gamma \sim \text{Gamma}\left(\frac{n}{2} - 1, \frac{A_\gamma}{2}\right).$$

4. Given σ and γ, sample

$$\beta \mid y, \sigma, \gamma \sim N_{m+1}\left(K_\gamma^{-1} X^T y, \sigma^2 K_\gamma^{-1}\right).$$

Thus the reparametrization $\tau = \sigma\gamma$ is useful because it turns the posterior calculation into three explicit steps: a one-dimensional posterior for γ, a Gamma conditional posterior for $1/\sigma^2$, and a Gaussian conditional posterior for β. "'

2.6 Induced Prior for the Regression Function

Our regression function is being modeled as:

$$f(x) = \beta_0 + \beta_1 x + \beta_2(x-1)_+ + \cdots + \beta_m(x-(m-1))_+ \tag{29}$$

where $\beta_0, \beta_1, \ldots, \beta_m$ are all independent with

$$\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C) \quad \text{and} \quad \beta_2, \ldots, \beta_m \overset{\text{i.i.d}}{\sim} N(0, \tau^2).$$

This can also be seen as a prior on the regression function f more directly. For every $0 \le u_1 < \cdots < u_k < \infty$, the joint distribution of $(f(u_1), \ldots, f(u_k))$ induced by (29) will be multivariate normal. This implies that $(f(u), u \ge 0)$ is a Gaussian Process. The description of the GP can be done in terms of the mean function $f(u)$ and the covariance kernel $\text{Cov}(f(u), f(v))$.

The mean function is clearly zero (because each $\beta_j = 0$). The covariance kernel is given by:

$$\begin{aligned}
&\text{cov}(f(u), f(v)) \\
&= \text{cov}\left(\beta_0 + \beta_1 u + \beta_2(u-1)_+ + \cdots + \beta_m(u-(m-1))_+, \beta_0 + \beta_1 v + \beta_2(v-1)_+ + \cdots + \beta_m(v-(m-1))_+\right) \\
&= C(1+uv) + \tau^2 \sum_{j=1}^{m-1} (u-j)_+(v-j)_+ \\
&= C(1+uv) + \tau^2 \sum_{j=1}^{k} (u-j)(v-j) \\
&= C(1+uv) + \tau^2 \left(uvk + \frac{1}{6}k(k+1)(2k+1) - (u+v)\frac{k(k+1)}{2}\right)
\end{aligned}$$

where $k = \lfloor u \wedge v \rfloor$ (and $u \wedge v := \min(u, v)$).

Suppose now we use the approximation

$$k \approx u \wedge v \quad \text{and} \quad k(k+1)(2k+1) \approx 2k^3 \approx 2(u \wedge v)^3 \quad \text{and} \quad k(k+1) \approx k^2 \approx (u \wedge v)^2.$$

Then the covariance kernel becomes:

$$\text{cov}(f(u), f(v)) \approx C(1+uv) + \tau^2 \left(uv(u \wedge v) + \frac{(u \wedge v)^3}{3} - \frac{(u \wedge v)^2}{2}(u+v)\right).$$

It turns out that the right hand side above is precisely the covariance kernel of:

$$G_x = \beta_0 + \beta_1 x + \tau I_x$$

where $\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C)$ and I_x is **Integrated Brownian Motion** on $[0, \infty)$. We will revisit this a little later in the course when we discuss nonlinear function/curve fitting.

2.7 Summary

In classical ridge regression, the regularization parameter λ is chosen using cross-validation. The Bayesian machinery provides a clean, principled and computationally amenable way to avoid cross validation to select tuning parameters.

Instead of directly penalizing the coefficients, we place a prior distribution on them. For a standard linear regression model

$$y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I), \tag{30}$$

the usual diffuse prior on β leads to the ordinary least squares estimator as the posterior mean. Thus ordinary least squares can itself be viewed as a Bayesian procedure corresponding to a nearly flat prior. To encourage smoother fits, we instead place a Gaussian shrinkage prior on the coefficients. Specifically, we assume that these coefficients have mean zero and a relatively small variance.

A standard conjugate Gaussian calculation shows that the posterior mean takes the form

$$E(\beta \mid y, \sigma, \tau) = \left(X^T X + \frac{1}{\gamma^2} J\right)^{-1} X^T y, \tag{31}$$

where $\gamma = \tau/\sigma$ and J is a diagonal matrix that penalizes only the coefficients corresponding to the ReLU basis functions.

This expression is identical to the ridge regression estimator

$$\widehat{\beta}_{\text{ridge}} = (X^T X + \lambda J)^{-1} X^T y \tag{32}$$

with

$$\lambda = \frac{1}{\gamma^2} = \frac{\sigma^2}{\tau^2}. \tag{33}$$

Thus ridge regression with penalty λ may be interpreted as a Bayesian posterior mean estimator under a Gaussian shrinkage prior for fixed τ and σ where $\lambda = \frac{\sigma^2}{\tau^2}$.

An important advantage of the Bayesian framework is that the amount of regularization need not be selected by cross-validation. Instead, one may place a prior on σ, γ (in turn producing a prior on λ) and estimate it from the data. The resulting posterior distribution provides not only a point estimate of the regression coefficients, but also a complete quantification of uncertainty.

$\square$

3 The Lasso

Ridge regression shrinks the regression coefficients toward zero and often leads to substantial improvements in prediction accuracy. However, ridge regression has one important limitation: it does not perform variable selection. Unless the tuning parameter is extremely large, all predictors remain in the model.

In many high-dimensional applications, we believe that only a small subset of the predictors are truly important. This motivates a different form of regularization that not only shrinks coefficients toward zero but can also set some coefficients exactly equal to zero.

The most widely used method of this type is the *Least Absolute Shrinkage and Selection Operator*, or simply the *lasso*.

The following slides from Hastie's lecture notes introduce the lasso, explain its variable selection property, and compare it with ridge regression.

The Lasso

- Ridge regression does have one obvious disadvantage: unlike subset selection, which will generally select models that involve just a subset of the variables, ridge regression will include all p predictors in the final model
- The *Lasso* is a relatively recent alternative to ridge regression that overcomes this disadvantage. The lasso coefficients, $\hat{\beta}^L_\lambda$, minimize the quantity

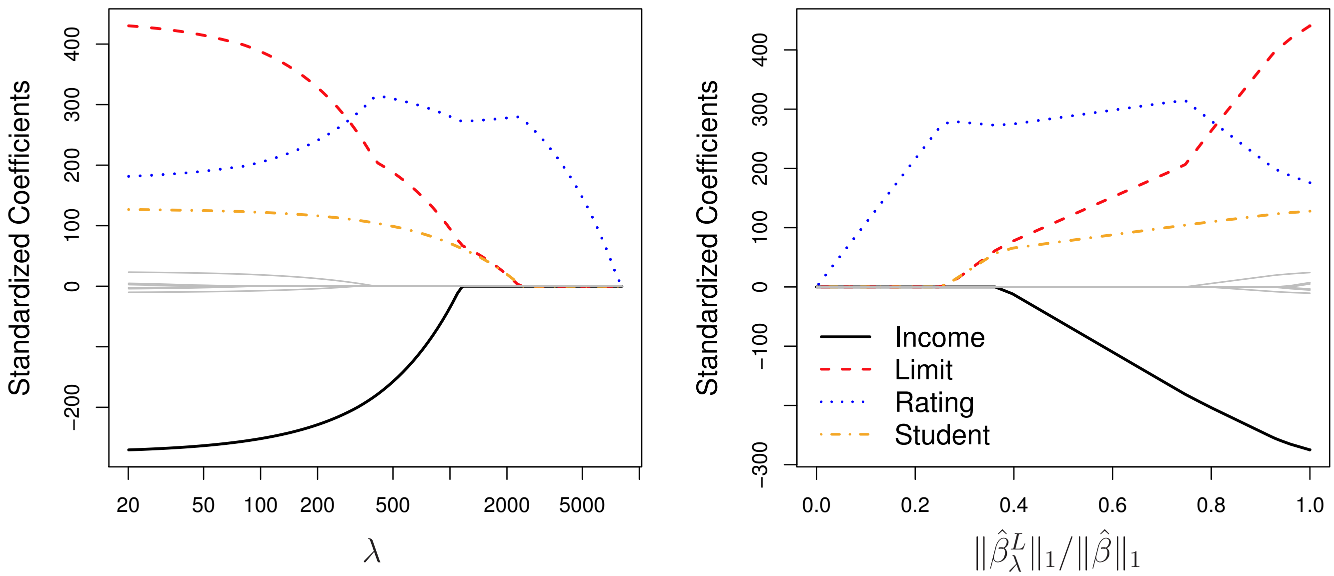
$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \text{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|.$$

- In statistical parlance, the lasso uses an ℓ_1 (pronounced "ell 1") penalty instead of an ℓ_2 penalty. The ℓ_1 norm of a coefficient vector β is given by $\|\beta\|_1 = \sum|\beta_j|$.

The Lasso: continued

- As with ridge regression, the lasso shrinks the coefficient estimates towards zero.
- However, in the case of the lasso, the ℓ_1 penalty has the effect of forcing some of the coefficient estimates to be exactly equal to zero when the tuning parameter λ is sufficiently large.
- Hence, much like best subset selection, the lasso performs *variable selection*.
- We say that the lasso yields *sparse* models — that is, models that involve only a subset of the variables.
- As in ridge regression, selecting a good value of λ for the lasso is critical; cross-validation is again the method of choice.

Example: Credit dataset



The Variable Selection Property of the Lasso

Why is it that the lasso, unlike ridge regression, results in coefficient estimates that are exactly equal to zero?

One can show that the lasso and ridge regression coefficient estimates solve the problems

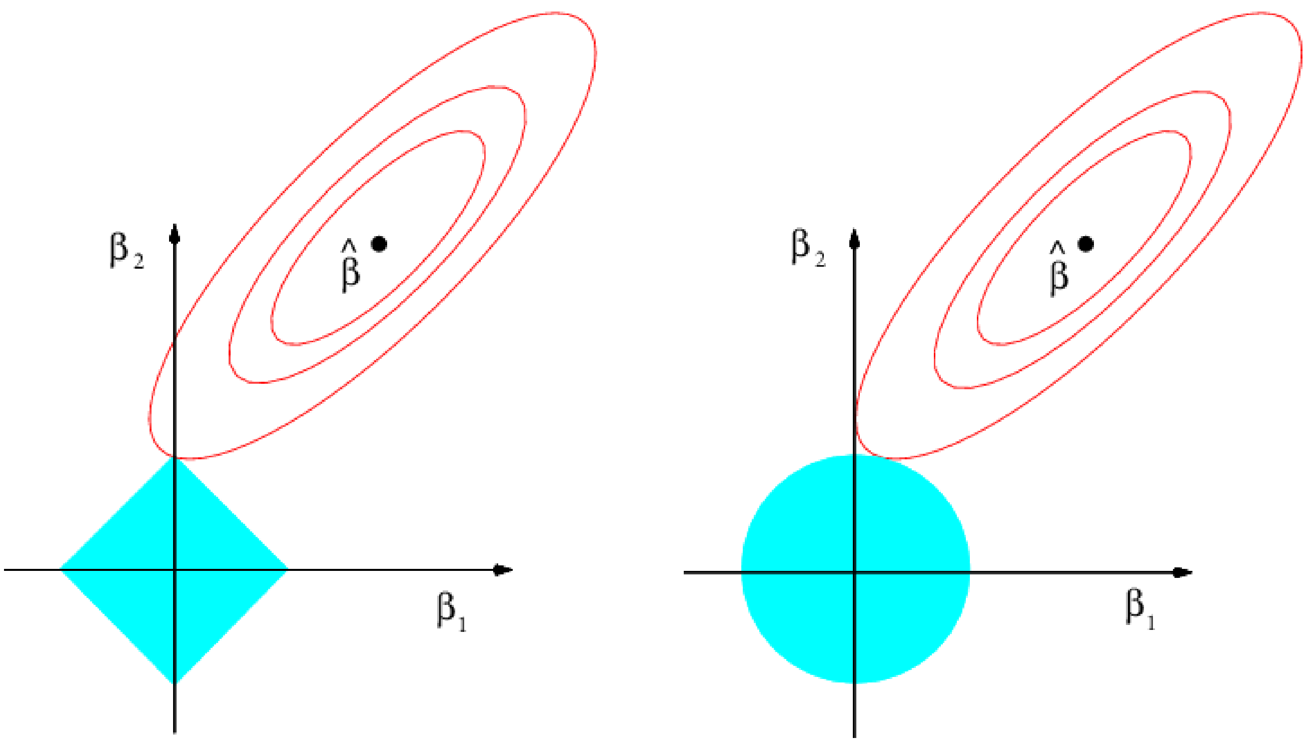
$$\underset{\beta}{\text{minimize}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq s$$

and

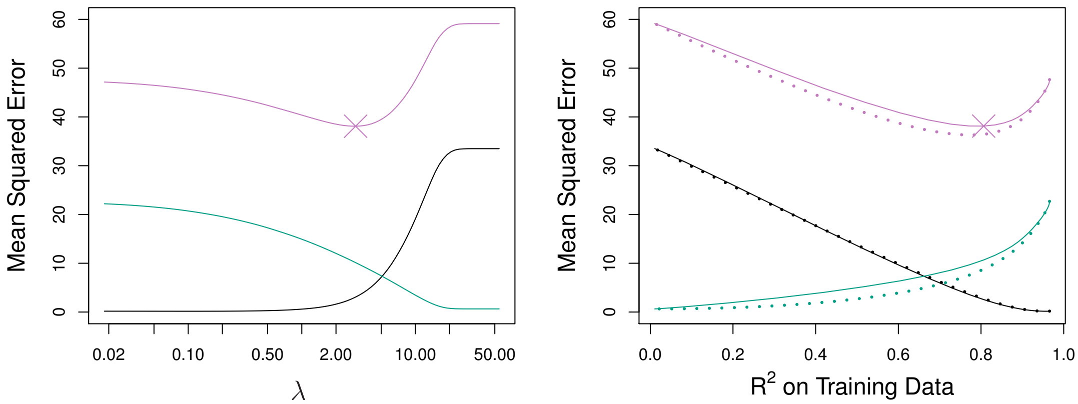
$$\underset{\beta}{\text{minimize}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 \leq s,$$

respectively.

The Lasso Picture

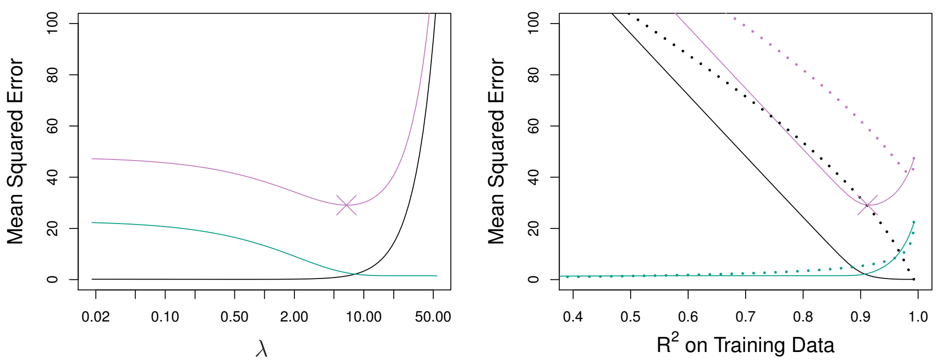


Comparing the Lasso and Ridge Regression



Left: *Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso on simulated data set of Slide 32.*
Right: *Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.*

Comparing the Lasso and Ridge Regression: continued



Left: *Plots of squared bias (black), variance (green), and test MSE (purple) for the lasso. The simulated data is similar to that in Slide 38, except that now only two predictors are related to the response.* *Right:* *Comparison of squared bias, variance and test MSE between lasso (solid) and ridge (dashed). Both are plotted against their R^2 on the training data, as a common form of indexing. The crosses in both plots indicate the lasso model for which the MSE is smallest.*

Conclusions

- These two examples illustrate that neither ridge regression nor the lasso will universally dominate the other.
- In general, one might expect the lasso to perform better when the response is a function of only a relatively small number of predictors.
- However, the number of predictors that is related to the response is never known *a priori* for real data sets.
- A technique such as cross-validation can be used in order to determine which approach is better on a particular data set.

4 Some Convex Analysis for the Lasso

Consider the lasso optimization problem

$$\hat{\beta} \in \arg\min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1 \right\}, \qquad \lambda > 0.$$

4.1 KKT Conditions

The lasso is a convex optimization problem. Therefore, the global optimum can be characterized exactly through Karush–Kuhn–Tucker (KKT) conditions. In this section we develop some of the basic convex analysis that underlies the lasso.

The lasso objective is convex, but it is not differentiable because of the absolute value penalty. We therefore use subgradients.

Definition 4.1 (Subgradient and subdifferential)**.** Let $f : \mathbb{R}^p \to \mathbb{R}$ be a convex function. A vector $g \in \mathbb{R}^p$ is called a *subgradient* of f at $\beta \in \mathbb{R}^p$ if

$$f(u) \geq f(\beta) + g^T(u - \beta) \qquad \text{for all } u \in \mathbb{R}^p.$$

The set of all subgradients of f at β is called the *subdifferential* of f at β, and is denoted by

$$\partial f(\beta) = \left\{ g \in \mathbb{R}^p : f(u) \geq f(\beta) + g^T(u - \beta) \text{ for all } u \in \mathbb{R}^p \right\}.$$

Theorem 4.2 (First-order optimality condition for convex functions)**.** *Let $f : \mathbb{R}^p \to \mathbb{R}$ be a convex function. Then $\widehat{\beta}$ minimizes f over $\mathbb{R}^p$ if and only if*

$$0 \in \partial f(\widehat{\beta}).$$

Proof. Suppose first that $0 \in \partial f(\widehat{\beta})$. By the definition of subgradient, for every $u \in \mathbb{R}^p$,

$$f(u) \geq f(\widehat{\beta}) + 0^T(u - \widehat{\beta}) = f(\widehat{\beta}).$$

Thus $\widehat{\beta}$ is a minimizer of f.

Conversely, suppose that $\widehat{\beta}$ minimizes f. Then for every $u \in \mathbb{R}^p$,

$$f(u) \geq f(\widehat{\beta}).$$

Equivalently,

$$f(u) \geq f(\widehat{\beta}) + 0^T(u - \widehat{\beta}) \qquad \text{for all } u \in \mathbb{R}^p.$$

Therefore 0 is a subgradient of f at $\widehat{\beta}$, that is,

$$0 \in \partial f(\widehat{\beta}).$$

□

Definition 4.3. The subgradient of the absolute value function is

$$\partial|t| = \begin{cases} \{1\}, & t > 0, \\ [-1, 1], & t = 0, \\ \{-1\}, & t < 0. \end{cases}$$

Consequently,

$$\partial\|\beta\|_1 = \{ s \in \mathbb{R}^p : s_j \in \partial|\beta_j|, \ j = 1, \ldots, p \}.$$

Theorem 4.4 (KKT condition for the lasso). *A vector $\hat{\beta}$ solves the lasso problem if and only if there exists $s \in \partial\|\hat{\beta}\|_1$ such that*

$$X^T(y - X\hat{\beta}) = \lambda s.$$

Equivalently,

$$s_j \in \begin{cases} \{1\}, & \hat{\beta}_j > 0, \\ [-1, 1], & \hat{\beta}_j = 0, \\ \{-1\}, & \hat{\beta}_j < 0. \end{cases}$$

Proof. A point minimizes a convex function if and only if zero belongs to its subdifferential. The gradient of the squared-error part is

$$\nabla_\beta \left\{ \frac{1}{2}\|y - X\beta\|_2^2 \right\} = -X^T(y - X\beta).$$

Therefore $\hat{\beta}$ is optimal if and only if

$$0 \in -X^T(y - X\hat{\beta}) + \lambda\, \partial\|\hat{\beta}\|_1.$$

This is equivalent to saying that there exists $s \in \partial\|\hat{\beta}\|_1$ such that

$$X^T(y - X\hat{\beta}) = \lambda s.$$

□

When $p > n$, the lasso objective is usually not strictly convex in β. Therefore the coefficient vector need not be unique. However, the fitted values are always unique.

Theorem 4.5 (Uniqueness of fitted values). *If $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ are two lasso solutions, then*

$$X\hat{\beta}^{(1)} = X\hat{\beta}^{(2)}.$$

Proof. Let

$$u_1 = X\hat{\beta}^{(1)}, \qquad u_2 = X\hat{\beta}^{(2)}.$$

Suppose, for contradiction, that $u_1 \neq u_2$.

Because

$$u \mapsto \frac{1}{2}\|y - u\|_2^2$$

is strictly convex, we have

$$\frac{1}{2}\left\| y - \frac{u_1 + u_2}{2} \right\|_2^2 < \frac{1}{2}\left[\frac{1}{2}\|y - u_1\|_2^2 + \frac{1}{2}\|y - u_2\|_2^2 \right].$$

On the other hand, by convexity of the ℓ_1-norm,

$$\left\| \frac{\hat{\beta}^{(1)} + \hat{\beta}^{(2)}}{2} \right\|_1 \leq \frac{1}{2}\|\hat{\beta}^{(1)}\|_1 + \frac{1}{2}\|\hat{\beta}^{(2)}\|_1.$$

Let

$$\bar{\beta} = \frac{\hat{\beta}^{(1)} + \hat{\beta}^{(2)}}{2}.$$

Then

$$X\bar{\beta} = \frac{u_1 + u_2}{2}.$$

Therefore

$$\begin{aligned} L(\bar{\beta}) &= \frac{1}{2}\|y - X\bar{\beta}\|_2^2 + \lambda\|\bar{\beta}\|_1 \\ &< \frac{1}{2}\left[\frac{1}{2}\|y - X\hat{\beta}^{(1)}\|_2^2 + \frac{1}{2}\|y - X\hat{\beta}^{(2)}\|_2^2\right] + \lambda\left[\frac{1}{2}\|\hat{\beta}^{(1)}\|_1 + \frac{1}{2}\|\hat{\beta}^{(2)}\|_1\right] \\ &= \frac{1}{2}L(\hat{\beta}^{(1)}) + \frac{1}{2}L(\hat{\beta}^{(2)}). \end{aligned}$$

Since both $\hat{\beta}^{(1)}$ and $\hat{\beta}^{(2)}$ are minimizers, the right-hand side is equal to the minimum lasso objective value. This is impossible, because $\bar{\beta}$ would then have strictly smaller objective value than the minimum.

Hence $u_1 = u_2$, and therefore

$$X\hat{\beta}^{(1)} = X\hat{\beta}^{(2)}.$$

□

Although the lasso coefficient vector may fail to be unique, different solutions cannot disagree in an arbitrary way.

Proposition 4.6 (Sign consistency). *Suppose $\hat{\beta}$ and $\tilde{\beta}$ are two lasso solutions. If*

$$\hat{\beta}_j \neq 0 \qquad \textit{and} \qquad \tilde{\beta}_j \neq 0,$$

then

$$\operatorname{sign}(\hat{\beta}_j) = \operatorname{sign}(\tilde{\beta}_j).$$

Proof. By the previous theorem,

$$X\hat{\beta} = X\tilde{\beta}.$$

Hence the residual vector

$$r = y - X\hat{\beta}$$

is the same for every lasso solution.

By the KKT condition,

$$s = \frac{1}{\lambda}X^T r$$

is also unique.

If $\hat{\beta}_j > 0$, then $s_j = 1$. If $\tilde{\beta}_j < 0$, then $s_j = -1$. This is impossible because s_j is unique. The same argument rules out the reverse possibility. Thus common nonzero coefficients must have the same sign. □

Remark 4.7. *The above is in contrast to OLS, where the signs of the solutions can change when $p > n$. Check this yourself by creating an example.*

4.2 Coordinate Descent for Computing the Lasso

Unlike ridge regression, the lasso does not have a closed-form solution. The most common practical algorithm for computing the lasso is coordinate descent.

The basic idea is simple: hold all coefficients fixed except one, optimize over that coefficient, and then cycle through the coordinates.

4.2.1 One-Dimensional Coordinate Update

Fix all coefficients except β_j. Define the partial residual

$$r_j = y - \sum_{k \neq j} X_k \beta_k. \tag{34}$$

Then the lasso objective, as a function of β_j alone, is

$$Q_j(\beta_j) = \frac{1}{2}\|r_j - X_j\beta_j\|_2^2 + \lambda|\beta_j|. \tag{35}$$

Expanding,

$$Q_j(\beta_j) = \frac{1}{2}X_j^T X_j \left(\beta_j - \frac{X_j^T r_j}{X_j^T X_j}\right)^2 + \lambda|\beta_j| + C, \tag{36}$$

where C does not depend on β_j.
Thus the coordinate update reduces to the one-dimensional optimization problem

$$\min_x \; c(x-a)^2 + \lambda|x|, \tag{37}$$

where

$$c = \frac{1}{2}X_j^T X_j, \qquad a = \frac{X_j^T r_j}{X_j^T X_j}. \tag{38}$$

The solution is

$$\hat{x} = S\left(a, \frac{\lambda}{2c}\right), \tag{39}$$

where

$$S(a,\gamma) = \operatorname{sign}(a)(|a| - \gamma)_+. \tag{40}$$

Exercise: Show this yourself!
Therefore,

$$\hat{\beta}_j = S\left(\frac{X_j^T r_j}{X_j^T X_j}, \frac{\lambda}{X_j^T X_j}\right). \tag{41}$$

Equivalently,

$$\hat{\beta}_j = \frac{1}{X_j^T X_j} S(X_j^T r_j, \lambda). \tag{42}$$

4.2.2 The Algorithm

The coordinate descent algorithm is as follows.

1. Initialize $\beta^{(0)}$, often at the zero vector.

2. For $j = 1, \ldots, p$, compute the partial residual
$$r_j = y - \sum_{k \neq j} X_k \beta_k.$$
3. Update
$$\beta_j \leftarrow \frac{1}{n} S(X_j^T r_j, \lambda).$$
4. Cycle through the coordinates until the objective value no longer changes appreciably.

The soft-thresholding formula explains why the lasso produces exact zeros. If
$$|X_j^T r_j| \leq \lambda,$$
then the update gives
$$\beta_j = 0.$$
Thus variables whose correlation with the current residual is too small are removed from the model.

4.2.3 Convergence of Coordinate Descent

Theorem 4.8 (Convergence of coordinate descent)**.** *[Tseng, 2001] For the lasso objective*
$$F(\beta) = \frac{1}{2}\|y - X\beta\|_2^2 + \lambda\|\beta\|_1,$$
cyclic coordinate descent converges to a global minimizer of F.

Proof idea. Each coordinate update exactly minimizes the objective over one coordinate while keeping the others fixed. Therefore the sequence of objective values is nonincreasing.

Since F is bounded below, the objective values converge.

This alone does not prove convergence of the iterates. Coordinate descent for the lasso is guaranteed to converge by a general theorem of Tseng [Tseng(2001)] on block coordinate descent methods for convex optimization. It states that for convex functions of the form
$$F(\beta) = g(\beta) + \sum_{j=1}^{p} h_j(\beta_j),$$
where g is continuously differentiable and convex and each h_j is convex, cyclic coordinate descent converges to a global minimizer under mild regularity conditions.

The lasso has exactly this form, with
$$g(\beta) = \frac{1}{2}\|y - X\beta\|_2^2$$
and
$$h_j(t) = \lambda|t|.$$
Therefore cyclic coordinate descent converges to a global lasso solution. □

Remark 4.9. *Coordinate descent is the basis of many practical lasso implementations, including the widely used `glmnet` package. In practice, the algorithm is made even faster by computing the solution over a decreasing sequence of λ-values and using the previous solution as a warm start. Although coordinate descent is a classical optimization method, its modern use for the lasso was popularized by the highly efficient `glmnet` algorithm of Friedman, Hastie and Tibshirani [Friedman et al.(2010)Friedman, Hastie, and Tibshirani]. The convergence theory follows from the more general results of Tseng [Tseng(2001)].*

5 Bayesian and Frequentist Perspectives on the Lasso

The lasso has a natural interpretation from both the Bayesian and frequentist points of view.

Bayesian interpretation. Consider the linear regression model

$$Y = X\beta + \varepsilon, \qquad \varepsilon \sim N(0, \sigma^2 I_n). \tag{43}$$

Suppose that the regression coefficients are assigned independent Laplace (double-exponential) priors

$$\beta_j \overset{\text{i.i.d.}}{\sim} \text{Laplace}(0, b), \tag{44}$$

whose density is

$$p(\beta_j) = \frac{1}{2b} \exp\left(-\frac{|\beta_j|}{b}\right). \tag{45}$$

The joint prior is therefore

$$p(\beta) \propto \exp\left(-\frac{1}{b}\sum_{j=1}^{p} |\beta_j|\right). \tag{46}$$

Combining this prior with the Gaussian likelihood, the negative log-posterior is

$$\frac{1}{2\sigma^2}\|y - X\beta\|_2^2 + \frac{1}{b}\sum_{j=1}^{p} |\beta_j| + C. \tag{47}$$

Hence the posterior mode (MAP estimator) is

$$\hat{\beta}_{\text{MAP}} = \arg\min_{\beta} \left\{ \frac{1}{2}\|y - X\beta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}, \tag{48}$$

where

$$\lambda = \frac{\sigma^2}{b}. \tag{49}$$

Thus, the lasso estimator is precisely the MAP estimator under independent Laplace priors on the regression coefficients.

The Laplace prior is sharply peaked at zero while having heavier tails than a Gaussian prior. The sharp peak encourages many coefficients to be shrunk toward zero, while the heavier tails allow genuinely large coefficients to be estimated with relatively little shrinkage.

It is important to distinguish between the lasso estimator and a fully Bayesian analysis. The lasso is the *posterior mode* (or MAP estimator) under the Laplace prior. In contrast, a fully Bayesian analysis would summarize the entire posterior distribution, for example by computing the posterior mean, credible intervals, or posterior samples. Unlike the MAP estimator, the posterior mean is generally *not* sparse, and samples drawn from the posterior distribution are also typically not sparse. Thus, the exact zeros produced by the lasso arise from maximizing the posterior density rather than from averaging over the posterior distribution.

Frequentist interpretation. From a frequentist perspective, the lasso is viewed as a penalized least-squares estimator. Unlike ordinary least squares, however, the sampling distribution of the lasso estimator is not available in closed form. The non-differentiability of the ℓ_1 penalty causes some coefficients to be estimated exactly as zero, making the estimator a non-smooth function of the data.

As a result, the sampling distribution of $\hat{\beta}$ is generally non-Gaussian and may even contain point masses at zero. I am not aware of a general sampling distribution result for lasso estimates which allow us to do valid inference.

References

[Friedman et al.(2010)Friedman, Hastie, and Tibshirani] Jerome Friedman, Trevor Hastie, and Rob Tibshirani. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1):1–22, 2010.

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